VIMA: RGB-Only Spatial Evidence Memory for Construction Video

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Abstract

Egocentric construction video is a poor substrate for direct visual-language-model (VLM) judgment. In our pilot data, frontier VLMs produced fluent but unsafe spatial claims: hallucinating site objects, mis-estimating scaffold distance, and treating visual activity as evidence without binding it to a frame, object, or location. We present VIMA, an RGB-only spatial evidence memory for hardhat footage. VIMA decomposes video understanding into specialist perception stages—object boxes, SAM masks, monocular depth, depth-delta frame filtering, 3D reconstruction, and episodic retrieval—before any VLM writes an answer. The core contribution is a depth-delta prefilter: consecutive frames whose normalized depth disagreement exceeds $\delta = 0.25$ are treated as over-motion pairs and withheld from reconstruction. In hackathon pilot runs, this filter removed 57% of wild construction frame pairs, improved MASt3R translational pose error by 59% on TUM `freiburg1_desk`, reduced Ironsite masonry translation norm by 70% (0.370 to 0.111), and produced a COLMAP reconstruction with 1,770 points, 19/31 registered images, and 1.199px reprojection error. VIMA's final artifact is not a caption but an auditable object-event memory: episode ids, evidence frames, tracks, depth facts, and retrieved answers. This reframes construction AI from "ask a VLM what happened" to "verify spatial claims against recoverable evidence."

1 Introduction

Construction progress and compliance are spatial problems. A useful system must answer not only *what* appears in a frame, but also *where* it is, whether it persists across time, which physical object changed, and whether a claim can be audited later. Egocentric-video benchmarks such as Ego4D show why memory and evidence retrieval matter in first-person video [2]. Raw hardhat footage makes this harder: fisheye distortion, motion blur, repeated materials, partial occlusion, scaffold geometry, and multiple workers make many single-frame judgments ambiguous.

Our initial strategy was VLM-first: give Gemini, Claude, or an open VLM construction frames and ask for safety or progress judgments. This failed in exactly the way that matters for a construction company. The answers were polished, but the spatial grounding was brittle. Gemini hallucinated absent objects such as rebar in masonry frames. Claude-style distance estimates on scaffold scenes were off by roughly 3× in pilot checks. These errors are not cosmetic. A false spatial claim could credit the wrong work, flag the wrong hazard, or produce an audit trail that cannot be defended.

Attempt	Observed failure
Raw VLM object judgment	Hallucinated construction objects in frame-level reasoning.
Raw VLM distance judgment	Scaffold distance estimates were off by about 3× in pilot checks.
Single-frame productivity label	Over-labeled activity as non-contributory in early CII runs.
VLM-only answer	Could answer fluently without citing frames, tracks, or geometry.

Table 1: Why VIMA moved away from direct VLM judgment. These are hackathon pilot observations used to choose the system design rather than final benchmark claims.

VIMA therefore treats the VLM as the final synthesizer, not the spatial reasoner. This follows the broader lesson of recent spatial VLM work: language models can benefit from explicit spatial structure, but the structure should be externalized rather than assumed [1]. The system first builds a structured memory from RGB video, then asks a VLM or local open model to answer from that memory.

2 Failure Mode

The task we target is **spatially verifiable work evidence**: given hardhat video, determine whether an activity claim is supported by frames, object tracks, depth/order evidence, and reconstruction. Humans can inspect the footage and say, for example, "masonry activity is visible near this block wall at these timestamps." VLMs can often describe the scene, but they do not naturally return the chain of evidence that makes the claim inspectable.

3 Method

VIMA is an RGB-only spatial evidence pipeline. The input is a hardhat video; the output is a queryable memory whose claims point back to frames and intermediate evidence.

3.1 Perception Stack

Frames are sampled from video and passed through four specialist stages:

1. **Object proposals**: YOLO/Codex-style labels for workers, scaffold, concrete block walls, guardrails, material stacks, tools, and open edges.
2. **Masks and tracks**: box-prompt SAM masks convert rectangles into object support regions and maintain object identities across frames.
3. **Monocular depth**: DepthAnythingV2-style relative depth estimates provide near/mid/far ordering for each masked object [7].
4. **Geometry**: MASt3R/COLMAP reconstructions estimate camera motion and sparse structure on filtered frame sets [3, 4, 5].

The key design decision is that every model output becomes evidence metadata rather than an immediate natural-language conclusion.

3.2 Depth-Delta Frame Filtering

Construction video contains many adjacent frames that are visually adjacent in time but poor for geometry: fast head turns, motion blur, and partial occlusion. We found that RGB similarity alone was a weak filter, so VIMA uses depth disagreement as a proxy for spatial instability.

For consecutive frames I_t and I_{t+1}, a monocular depth model produces normalized maps $D_t, D_{t+1} \in [0,1]^{H \times W}$. We compute

$$\Delta_D(t) = \sqrt{\frac{1}{HW} \sum_{x,y} (D_t(x,y) - D_{t+1}(x,y))^2}. \quad (1)$$

If $\Delta_D(t) > \delta$ with $\delta = 0.25$, the pair is marked as over-motion and skipped before reconstruction. This is intentionally simple: depth is not used as metric truth, but as an RGB-derived spatial oracle for whether two frames are likely to form a stable geometric pair.

3.3 Object-Event Episodic Memory

After masks and depth, VIMA compiles object-level facts into episodes:

- `masonry_work_candidate`
- `safety_edge_context`
- `scaffold_zone_visible`
- `foreground_worker_present`
- `material_staging_visible`

Each episode stores start/end time, evidence frames, involved tracks, spatial relations, depth facts, confidence, and retrieval text. The VLM answer layer receives this compact memory, not the full raw clip.

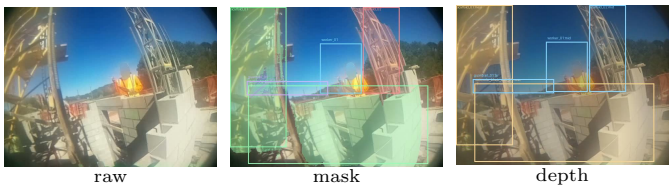

Figure 1: VIMA evidence views on an Ironsite masonry clip. The dashboard exposes raw, mask, and depth-conditioned views so an internal reviewer can inspect the claim path.

Component	Metric	Pilot result
Depth-delta filter	Dropped pairs	57%
MASt3R on TUM	RPE_t reduction	59%
Ironsite masonry	Translation norm	$0.370 \rightarrow 0.111$
Ironsite masonry	Translation improvement	70%
Ironsite masonry	Confidence change	+32%
COLMAP Ironsite	Registered images	19/31
COLMAP Ironsite	Sparse points	1,770
COLMAP Ironsite	Reprojection error	1.199px

Table 2: Preliminary geometry results from the hackathon pipeline. The TUM result used `freiburg1_desk`; Ironsite results used a masonry hardhat clip.

4 Pilot Results

We report preliminary hackathon results, not a full benchmark. The important pattern is that geometry improves when high-depth-disagreement frame pairs are filtered before reconstruction, and the final memory makes VLM answers auditable.

5 Answer Layer

VIMA supports both API VLMs and local open-weight probes. Gemini produced cleaner natural-language synthesis from the memory artifact when quota was available. Qwen2-VL-2B ran locally and confirmed that retrieved evidence frames and memory can be consumed without a remote API, though it followed citation instructions less reliably [6]. In both cases, the answer JSON stores retrieved episode ids and evidence frame names, so the response remains auditable.

For example, the query *"Was there masonry work happening near the wall?"* retrieved `episode_001` and `episode_007`; each contains concrete block wall tracks, worker tracks, proximity relations, depth facts, and supporting frame names.

6 Dashboard and Deployment

The dashboard is an internal evidence-review console rather than a consumer rewards app. It shows the current claim, the raw/mask/depth evidence views, run statistics, and an object-event timeline. Separately, a

Artifact	Value	Role
Sampled frames	10	Demo subset
Frame events	35	Object/depth facts
Episodes	10	Retrieval units
Top confidence	0.875	Masonry candidate
Qwen local answer	Success	Open VLM probe
Gemini answer	Success, quota-limited	Synthesis probe

Table 3: Object-event memory artifacts used by the dashboard and answer layer.

FastAPI backend was prototyped for bounty-style workflows with multi-witness verification: two independent witnesses must agree on the same frame before a payout-style action is enabled. For the construction-company dashboard, the same idea becomes an anti-larp audit rule: do not accept unsupported text; require evidence frames and independent signals.

7 Limitations

The current system is a hackathon prototype. The reported numbers come from small pilot runs and should be treated as directional until repeated on a frozen dataset. DepthAnythingV2 is relative, not metric, so VIMA uses it for ordering, filtering, and stability checks rather than absolute measurements. The CII classifier runs were inconsistent: one full run over-labeled all frames as non-contributory, while smaller temporal batches produced more plausible P/C/NC splits. This argues for temporal batching and human-reviewed benchmark labels before any safety-critical deployment.

8 Conclusion

VIMA's central claim is that construction VLM systems need a spatial evidence layer before language synthesis. Direct VLM prompting is too brittle for jobsite claims because it can hallucinate objects, mis-estimate distance, and omit the evidence path. By combining RGB-derived depth, frame filtering, object masks, reconstruction, and episodic memory, VIMA turns hardhat video into auditable spatial claims. The most promising technical contribution is depth-delta filtering: a simple monocular-depth check that removes unstable frame pairs before geometry and improves downstream reconstruction quality in pilot runs.

References

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